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Transparent, high strain point spinel glass-ceramics

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Abstract

Transparent, refractory glass-ceramics based on 10 nm crystals of spinel solid solution in a highly siliceous residual glass can be produced from compositions in the $\text{SiO}_2\text{--Al}_2\text{O}_3\text{--ZnO--MgO--TiO}_2\text{--ZrO}_2$ system. The glass-ceramics possess excellent chemical durability as well as thermal expansion coefficients of $35\text{--}40 \times 10^{-7}/^\circ\text{C}$ and strain point temperatures of over 900°C . Titania serves as both an extremely effective nucleating agent in these glasses and as an integral component of the spinel crystals. The materials are designed such that fluxes in the glass partition into the spinel crystals during the crystallization process, leaving a continuous residual glass very high in silica. The strain points of the resultant glass-ceramics approach those of vitreous silica and quartz materials which require significantly more expensive manufacturing processes. These glass-ceramics are excellent substrate candidates for high temperature, high quality polysilicon thin films. Potential products include solar cells and active matrix liquid crystal displays. © 1999 Elsevier Science B.V. All rights reserved.

1. Introduction

Transparent materials capable of withstanding high temperatures are valuable in numerous familiar applications such as woodstove windows and lamp envelopes. For many applications, however, high temperature capability is required not by the final product but rather by the intermediate processing necessary to build the glass into a final product. A common example of such processing is the deposition of semiconductor thin films on glass substrates during the manufacture of computer display screens. Thin film transistors of polycrystalline silicon (poly-Si TFTs) are widely viewed as one of the most promising technologies

enabling high density, high definition flat panel and liquid crystal displays [1–4]. Poly-Si allows the peripheral drive circuitry to be integrated onto the substrate, significantly reducing costs and increasing product reliability.

Most current poly-Si TFT device processing is carried out at low temperatures ($450\text{--}600^\circ\text{C}$) that are compatible with commercial flat panel display glasses, all of which have strain points lower than 700°C . Such low-temperature fabrication, however, requires the use of complex processes, such as the laser recrystallization of amorphous silicon, that are difficult to scale up to large area displays with the necessary precision and uniformity [1]. The highest quality poly-Si TFTs are fabricated at temperatures over 900°C ; such processes yield high electron mobilities for rapid switching and excellent TFT uniformity across large areas. While vitreous silica (fused quartz) substrates are

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currently employed as high-temperature poly-Si substrates [5,6], particularly for projection LCD systems, the combination of their high cost, size limitation, and thermal expansion mismatch to silicon limits their use. While some higher-temperature glasses have been developed with strain points between 800°C and 830°C (e.g. [7,8]), conventional multi-component glasses that are readily melted and formed do not possess the thermal robustness required for extensive 900°C processing.

Glass-ceramic materials can provide the ease of manufacture of a multi-component glass along with the advantage of being able to remove the fluxes from the glass, via crystallization, leaving a refractory 'glass' that otherwise would be incompatible with cost-effective manufacture. One such glass-ceramic family is based on spinel crystals in the system $\text{SiO}_2\text{--Al}_2\text{O}_3\text{--ZnO--MgO--TiO}_2\text{--ZrO}_2$.

2. Experimental procedures

Calculated amounts of sand, alumina, zinc oxide, magnesium oxide, titania, zircon, and, in some glasses, barium or cesium carbonate, were batched to yield about 1000 g of glass. Arsenic oxide and ammonium nitrate were added to fine the glass and to keep a high ratio of Ti^{4+} to Ti^{3+} in the glass and glass-ceramic, thereby minimizing any color. The materials were dry ball milled for 1 h and transferred to platinum crucibles. The batches were melted in electric furnaces for 6 h at temperatures of 1575–1625°C, cast into patties about 1 cm thick, and annealed at 725°C. Ceramming was carried out in small electric furnaces. Glass samples cut from the patties were held at 800°C for 1–2 h and at 900–1050°C for 2–4 h to complete crystallization.

Phase assemblages were identified using X-ray powder diffraction techniques (Phillips XRG 3100, Cu K α radiation). Mechanical and thermal properties of the glass-ceramic samples were measured using standard techniques. Young's modulus (room temperature sonic resonance) was measured according to ASTM C623-92. Modulus of rupture (MOR) was measured on abraded bars, $2 \times 1/4 \times 1/8$ in., using a four-point bend setup with a Sintech 2/G unit. Densities were measured by the buoy-

ancy method (ASTM C693). Thermal expansion coefficients were measured using the vitreous silica dilatometer measurement and thermal expansion curves measured via the dual push rod technique, both according to ASTM E228. Anneal and strain points of the glass-ceramic samples were measured by beam bending techniques according to ASTM C598.

Images of the crystalline microstructures were obtained with a JEOL 2000FX scanning transmission electron microscope on ion-milled thin sections. Atomic force microscopy (AFM) images of polished glass-ceramic surfaces were collected on a Digital Instruments Nanoscope III SPM, using contact mode.

3. Results

3.1. Composition, phase assemblage, and microstructure

Transparent glass-ceramics with high strain points and thermal expansion characteristics similar to those of silicon were obtained from the general composition area SiO_2 50–62 wt%, Al_2O_3 17–22%, ZnO 6–15%, MgO 0–5%, TiO_2 0–10%, ZrO_2 0–8%, BaO 0–4%, and Cs_2O 0–6%. Several compositions with thermal expansions closely matched to that of silicon are listed in Table 1.

The predominant and usually sole crystal phase in these materials consists of spinel solid solutions whose compositions lie within the space represented by $\text{MgAl}_2\text{O}_4\text{--ZnAl}_2\text{O}_4\text{--Mg}_2\text{TiO}_4\text{--Zn}_2\text{TiO}_4$. Spinel has cubic crystal structures and the general chemical formula AB_2O_4 , where A is a tetrahedrally coordinated, typically divalent metal such as Zn, Mg, or Fe, and B is an octahedrally coordinated, usually trivalent metal such as Al or Fe. The spinel crystals, about 30% by volume, are distributed uniformly throughout a matrix of siliceous residual glass and may be accompanied by minor amounts of tetragonal zirconia when ZrO_2 is used for nucleation. The spinel crystals range in size from 7.5 to 15 nm (75–150 Å), with an average diameter of 10 nm. A transmission electron micrograph of a spinel glass-ceramic microstructure is given as Fig. 1.

Table 1
Glass-ceramic compositions and properties

Oxide (wt%)	GC1	GC2	GC3	GC4	GC5	GC6
SiO ₂	58.3	57.0	55.0	57.0	57.3	59.3
Al ₂ O ₃	20.2	18.0	21.0	21.8	18.5	19.1
ZnO	8.4	14.0	13.5	14.0	11.6	9.0
MgO	4.2	—	—	—	2.4	2.5
Cs ₂ O	—	4.0	4.0	—	—	—
BaO	—	—	—	2.1	2.0	2.1
ZrO ₂	5.0	7.0	—	—	3.0	3.0
TiO ₂	3.0	—	5.5	5.2	5.0	5.0
(RO + R ₂ O)/Al ₂ O ₃	1.1	1.1	0.9	0.9	1.2	1.0
CTE ($\times 10^{-7}/^{\circ}\text{C}$) (25–300°C)	38	37	37	37	38	37
Strain point (°C)	945	928	934	935	922	910

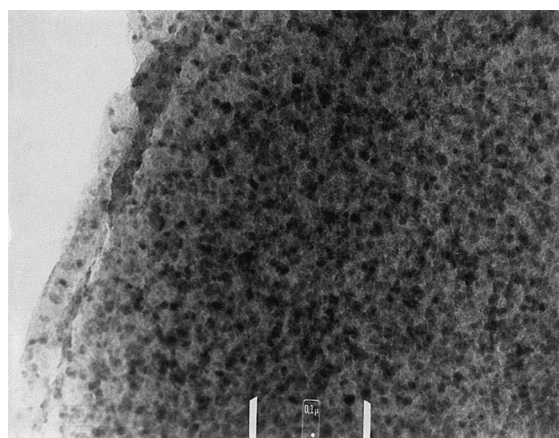


Fig. 1. STEM micrograph of microstructure of spinel glass-ceramic GC1, heat treated at 800°C-1 h, 900°C-2 h. Scale bar indicates 0.1 μm .

While zirconia alone, titania alone, or mixtures of the two oxides will nucleate the spinel phase [9–11], nucleation with zirconia alone is generally not desired from a practical point of view. Zirconia increases the density of the glass-ceramics, which is undesirable for most applications. Perhaps more important, previous unpublished work has shown that ZrO₂ significantly raises the liquidus temperature of these glasses, produces very steep high-temperature viscosity curves, and moreover is less efficient than titania in nucleating spinel in glasses that contain significant levels of magnesia.

The more magnesia in a composition, the higher the level of titania required to efficiently nucleate the spinel phase. While 5% TiO₂ is sufficient in

Mg-free spinel compositions, a minimum of 7.5% TiO₂ generally is required in ZrO₂-free compositions with more than about 2% MgO by weight. If not well nucleated, magnesia-containing glasses in this composition range tend to produce β -quartz solid solution and Mg-petalite instead of or in addition to spinel [12,13], leading to undesirable crystal growth and subsequent haziness or opacity in the glass-ceramic, as well as cracking in the most severe cases.

Titania serves as both an extremely effective nucleating agent in these glasses and as an integral component of the spinel crystals. While zirconia acts as a nucleating agent in the classical sense, whereby precipitated crystals of tetragonal ZrO₂ serve as the nucleant for zinc-rich spinel crystals [9], no evidence of a crystalline titanate phase is observed in TiO₂-nucleated glass-ceramics, even with TiO₂ levels as high as 9 wt%. Instead, nucleation is preceded by a highly uniform, ultra-fine-scale amorphous phase separation into silica-rich and titania/alumina-rich areas (Fig. 2). The spinel crystals then crystallize and grow within the latter globular-shaped areas. The absence of titanate phases in these glass-ceramics indicates the marked partitioning of Ti into the spinel crystal. The Ti⁴⁺ ion can enter the octahedral site of spinel by means of a charge balanced substitution such as Ti⁴⁺ + (Zn, Mg)²⁺ = 2 Al³⁺. It also is possible that the regular octahedral site of the spinel structure helps to stabilize the Ti³⁺ ion, which would provide another means of charge balance: a straight octahedral substitution of Ti³⁺ for Al³⁺. The possible stabilization of Ti³⁺ in spinel is supported

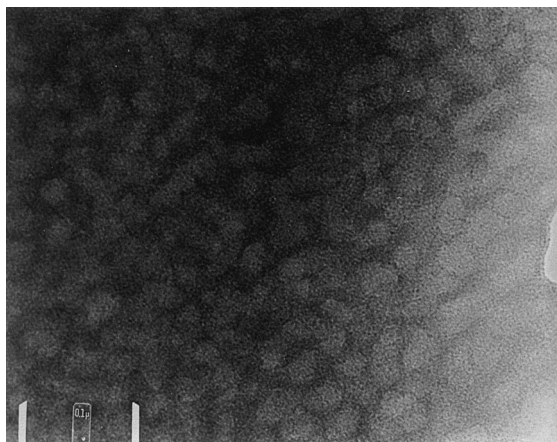


Fig. 2. STEM micrograph of phase separation in 600°C annealed glass of composition GC6. Scale bar indicates 0.1 μm .

by the observation that the gray color that is characteristic of Ti^{3+} occurs only in the glass-ceramic (in the absence of oxidizing agents such as arsenic oxide) and not in the precursor glass. Titaniferous spinels are common in rocks, where Fe–Ti-spinels are key components of the Fe–Ti oxides that serve as sensitive geothermometers and oxygen barometers [14].

3.2. Glass-ceramic properties

Table 2 compares properties of spinel glass-ceramic GC1 with those of vitreous silica [15,16] and of a commercial flat panel display glass, Corning Code 1737, a borosiluminosilicate glass that is expansion-matched to silicon [17].

The glass-ceramics possess thermal expansion coefficients of $35\text{--}40 \times 10^{-7}/^\circ\text{C}$. Fig. 3 shows expansion coefficient curves of glass-ceramic GC1 and of silicon [18]. The curves match very well. The difference in coefficients between the two materials is about 1% at 300°C (37.4 and $37.8 \times 10^{-7}/^\circ\text{C}$ for the glass-ceramic and silicon, respectively) and about 5% at 500°C. Curves for vitreous quartz [19] and Corning Code 1737 glass [17] are included for comparison.

Strain points of the glass-ceramics generally lie between 900°C and 940°C. The materials display excellent thermal stability, undergoing less than 10 ppm shrinkage in a simulated poly-Si thermal process of 900°C for 6 h, equivalent to the shrinkage obtained with fused silica.

The glasses undergo uniform densification of 3–4% on ceramming. Glass-ceramic densities range from 2.70 to 2.85 g/cm^3 and their Young's modulus values range from 90–110 GPa ($13\text{--}16 \times 10^6$ psi). Hard spinel crystals provide these materials with fairly high surface hardness. (Natural spinels have Mohs hardness values of 7.5–8.0 and occur frequently with corundum in emery deposits.) Typical Knoop hardness values (KHN/100 g load) are 550 for the precursor glass and 650 for the glass-ceramic. The glass-ceramics have mechanical strengths higher than those of glass and similar to those of common β -quartz solid solution glass-ceramics such as Visions[®], with modulus of rupture values on the order of 70–82 MPa ($10\text{--}12 \times 10^3$ psi).

Spinel glass-ceramics compare favorably to fused silica in overall durability, a result of their microstructure of stable oxide crystals in a matrix

Table 2
Comparison of properties of GC1 with vitreous silica and Corning Code 1737

	GC1	Vitreous Silica	Code 1737
CTE ($\times 10^{-7}/^\circ\text{C}$)	37	6	37.6
Strain point ($^\circ\text{C}$)	945	990–1090 ^a	666
Density (g/cm^3)	2.75	2.20	2.54
Young's modulus (GPa)	98	72	73
Knoop Hardness (KHN ₁₀₀)	650	500	460
Index of refraction	1.56	1.46	1.52
MOR (MPa)	73	60	56
Dielectric constant (25°C, 1 kHz)	5.5	3.8	5.7

^a The strain point of vapor phase deposition-fabricated fused silica (Corning Code 7940) is 990°C, while that of fused quartz (e.g. GE Type 124) is 1090°C.

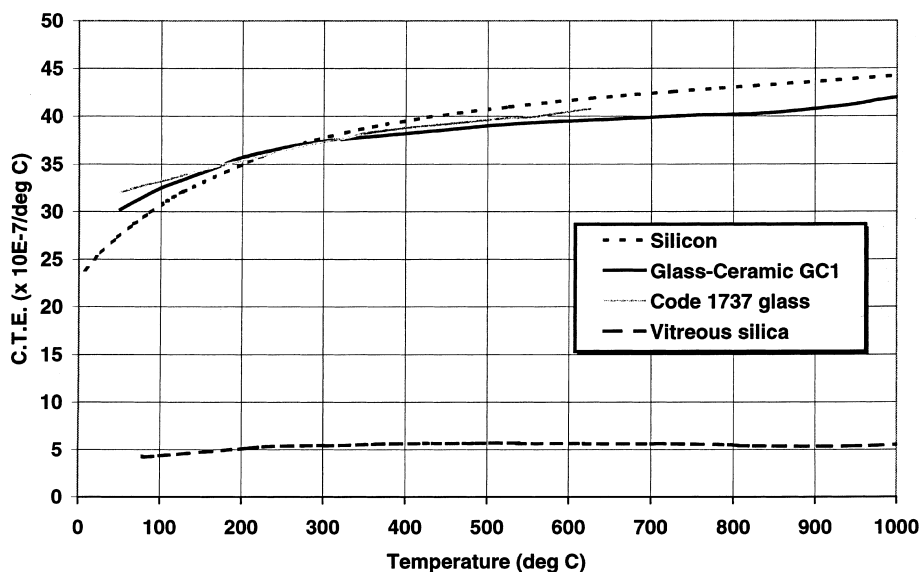


Fig. 3. Composite thermal expansion curves of spinel glass-ceramic GC1, silicon, fused quartz, and Corning Code 1737 glass.

of highly siliceous glass. Table 3 lists weight losses of these materials in standard acid, base, and buffered solutions.

The glass-ceramics range in color from water white (zirconia-nucleated) to very pale straw yellow (titania-nucleated), given the addition of 0.5 wt% As_2O_5 to the batch. Transmittance curves for a typical arsenic-fined, titania-nucleated spinel glass-ceramic and for fused silica and Code 1737 glass are given in Fig. 4. The yellow tint of the titania-nucleated materials is only slightly visible to the eye in 1 mm thick samples but is apparent in the transmittance cutoff in the near ultraviolet portion of the spectrum.

The ultra-fine crystal size allows these materials to be polished in the glass state and then ceramised without any significant increase in surface roughness, as measured by atomic force micros-

copy (AFM). Average roughness (R_a) values of polished surfaces typically range from 0.3–0.6 nm.

4. Discussion

Glass-ceramics typically have strain point temperatures significantly higher than those of their precursor glasses because the crystal phases take up many of the fluxing agents in the glass, leaving the residual glass with fewer fluxes – and therefore ‘harder’ – than the precursor glass. The spinel glass-ceramics are designed to have a residual glass that is very high in silica (>90%), thereby providing strain points well in excess of 850°C. This concept is aptly illustrated by comparing the properties of the GC1 glass-ceramic with those of its precursor glass.

Table 3
Chemical durability (weight loss) of spinel glass-ceramics and vitreous silica

Δ weight (mg/cm^2) in	GC2	GC3	GC6	Vitreous Silica
5% HCl, 95°C, 24 h	–0.002	–0.002	–0.021	–0.001
5% NaOH, 95°C, 6 h	–1.03	–0.76	–0.46	–0.57
10% HF, 25°C, 20 min	–0.79	–0.33	–0.19	–0.22
10% NH_4HF , 25°C, 20 min	–0.43	–0.23	–0.14	–0.20

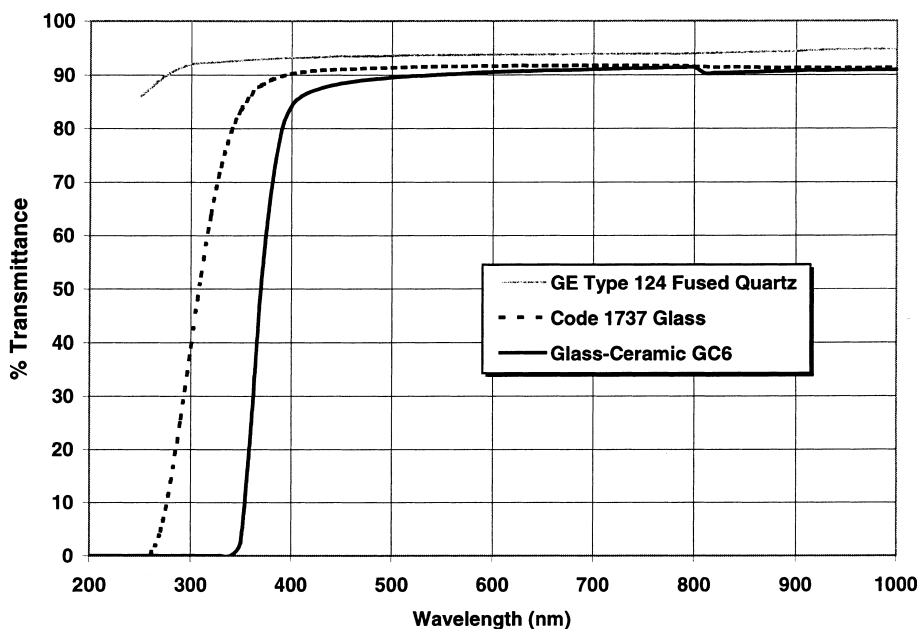


Fig. 4. Transmittance values through 1.0 mm thick samples of spinel glass-ceramic GC6, Corning Code 1737 glass, and GE Type 124 fused quartz.

	Precursor glass	Glass-ceramic
CTE	$28 \times 10^{-7}/^{\circ}\text{C}$	$37 \times 10^{-7}/^{\circ}\text{C}$
Strain point	730°C	945°C

The effect of crystallization of spinel is readily apparent. The strain point of the residual glass increases dramatically as its fluxes are removed, and the overall bulk expansion increases due to the high expansion of the spinel phase. (For example, MgAl_2O_4 crystals have a CTE of $\sim 75 \times 10^{-7}/^{\circ}\text{C}$.)

Silicate glass-ceramics with the highest strain points have residual glasses that resemble pure silica in structure and composition, with as few non-bridging oxygen atoms as possible. (Pure fused quartzes and silicas have strain points of 990 – 1100°C .) For this reason these glass-ceramics are designed to have molar ratios of $(\text{RO} + \text{R}_2\text{O})/\text{Al}_2\text{O}_3$ fairly close to unity (Table 1). While such ratios are less straightforward to interpret in multiphase glass-ceramic systems than they are in amorphous glasses, they can serve as useful indicators of the overall peraluminousness or peralkalinity of the system (excess alumina or alkali-alkaline earth, respectively), which in turn acts as a reasonable predictor of properties such as

strain point. The higher these ratios, the more non-bridging oxygen atoms there are in the residual glass and the lower the strain point of the glass-ceramic.¹ The strain points of the spinel glass-ceramics can range from as low as 850°C for relatively peralkaline glass-ceramics to over 950°C for compositions consisting basically of spinel components plus silica. The more peraluminous compositions commonly yield the highest strain points but poorer glass stability and liquidus/viscosity relationships.

5. Conclusions

Transparent, refractory glass-ceramics based on nanocrystals of spinel solid solution possess ther-

¹ The ratios remain reasonably valid in these glass-ceramics because the composition of the spinel phase can be approximated as a 1:1 molar ratio of $(\text{ZnO} + \text{MgO})$ to Al_2O_3 , ignoring TiO_2 . The large cations Ba^{2+} and Cs^{+} remain in the residual glass and must be compensated for with Al_2O_3 in order to minimize the number of non-bridging oxygen atoms.

mal expansion coefficients of $35\text{--}40 \times 10^{-7}/^{\circ}\text{C}$, matching that of silicon, and strain point temperatures over 900°C . The materials are designed such that fluxes in the glass partition into the spinel crystals during the crystallization process, leaving a continuous residual glass very high in silica. The strain points of the resultant glass-ceramics approach those of vitreous silica and quartz materials which require significantly more expensive manufacturing processes. The glass-ceramics have high elastic modulus values of 90–110 GPa and possess excellent chemical durability and thermal stability, maintaining transparency and dimensional stability for many hours at 900°C . The higher rigidity and strength of the glass-ceramics compared with vitreous silica and conventional glasses would permit the use of thinner sheets of the glass-ceramic, providing a potential weight advantage for display applications. These materials are excellent substrate candidates for high temperature, high quality polysilicon thin films, with potential products including active matrix liquid crystal displays and solar cells.

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